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Brief Communication

SMART-CYPS: an intelligent internet of things and machine learning powered crop yield prediction system for food security

Martin Kuradusenge[1] · Eric Hitimana[1] · Kambombo Mtonga[2] · Antoine Gatera[1] · Joseph Habiyaremye[3] · Jackson Ngabonziza[4] · Damien Hanyurwimfura[1] · Placide Rukundo[5] · Angelique Mukasine[1]

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Abstract
The sub-Saharan Africa region continues to experience food insecurity, a consequence of the less productive agricultural sector that has dragged to adapt to the effects of climate change. As the region's population continues to grow, there is a need to modernize the region's agricultural sector to meet the increasing food demand. Although extreme atmospheric conditions cannot be entirely mitigated, however, the integration of technologies such as the Internet of things (IoT) and machine learning (ML) can increase the quantity and quality of production from the crop fields. These technologies have the potential to empower agricultural management systems to handle both climatic and farm data in an orchestrated manner, and inform the formulation of effective strategies. This study presents the design and development of a system for predicting crop yields that integrates IoT and ML. The system combines historic and current weather and crop yield data to predict seasonal crop yields. The weather parameters including, rainfall, temperature, humidity and soil moisture are collected by IoT sensors and transmitted to the cloud for crop yield forecasting. The system is used to analyze seasonal yields of Irish-Potato and Maize in Musanze District of Rwanda. Using data over different agricultural seasons, the system achieved favorable predictive accuracy with mean absolute percentage error (MAPE) values of 0.339, 0.309, and 0.177 for two seasons of Irish potatoes and one season of maize, respectively. Such predictive yield systems can reduce food insecurity risks and enhance harvest efficiency by enabling early awareness of crop production, fostering effective strategies shared among decision-makers and stakeholders. While maize and Irish potatoes were the initial case studies, expansion to include other crops and more variables is envisioned.

Keywords IoT · Machine learning · Prediction · Weather · Climate change · Crop yield · Maize · Irish potatoes

1 Introduction

According to the World Bank, the number of people facing food insecurity increased from 135 to 345 million from 2019 to June 2022, due to the Ukraine war and the economic fallout from the COVID-19 pandemic that caused food prices to rise extremely [1]. Food insecurity is a result of several causes, including weather-related hazards, a lack of input resources, crop diseases, regional or international wars, and threats from both intentional and inadvertent activities [2, 3]. Weather-related threats include, high temperatures, water shortages, floods, droughts, and higher atmospheric CO_2 concentrations. For instance, corn and wheat production have gone down recently because of bad

✉ Martin Kuradusenge, m.kuradusenge@ur.ac.rw | [1]School of ICT, College of Science and Technology, University of Rwanda, Kigali, Rwanda. [2]Department of ICT, Rwanda Polytechnic, Gasabo, Kigali, Rwanda. [3]Narada Electronics, KN 84 St, Kigali, Rwanda. [4]Bank of Kigali Plc, Kigali, Rwanda. [5]International Potato Center (CIP), AfricaRice, Avaradrova, Ambovombe Androy, Madagascar.

weather, plant diseases, and a lack of water around the world [4]. Declining crop yields, particularly in the areas with the greatest food insecurity on the globe, will cause more people to live in poverty if there are no remedies. With climate change, more extreme weather occurrences are anticipated, which could paralyze agricultural productivity [5]. It is projected that by 2030, 43 million people in Africa alone will be living in poverty [1].

On a global scale, investments in technology development, adaption, and transfer might significantly lessen these effects [6]. To cope with the issue of food insecurity, the government of Rwanda has established different strategies including the Crop Intensification Programme (CIP), Livestock Intensification Program (LIP), Land Use Consolidation Program (LUCP), and others [7]. The CIP and LUCP among others were prioritized programs focusing on specialized crops with a target of increasing the agricultural production and food security in Rwanda [8, 9]. However, because most of the farming activities are predominately rainfed, Rwandan agriculture is extremely vulnerable to climate change and extreme weather events like droughts, floods, and severe storms [10]. To counter the impact of extreme weather events on food production, the Rwanda government has put in place different measures such as the promotion of drought-relieving irrigation methods. Under this initiative, local communities receive training in low-cost irrigation techniques such as rainwater collection, water storage, water pumping from water bodies, drought-resistant seeds, and food storage, where communities preserve extra harvest that can be used during drought periods [7]. However, rainfall deficits and rainfall induced disasters strike different regions in Rwanda leading to food insecurity being experienced in multiple regions of the country. For instance, eastern province is vulnerable to droughts due to irregular rainfall and long dry season that result in the shortage of food [11, 12]. On the other hand, the hilly and mountainous regions of Rwanda (namely north and west) are characterized by plenty of rain especially during the March—May rain season. As a result, floods and landslides are frequent, leading to poor crop harvest [13].

Even though it would be difficult to fully control climate conditions, technology use can reduce the magnitude of associated risks including food crisis. Historical weather and crop yields data can be used to predict the future production levels of different crops. This knowledge can be useful to policy makers enabling them to set some strategies aiming at reducing the risks of food insecurity. At present, fully documented studies on the prediction of crop production using combined weather and crop yield data are scarce in Rwanda. Also, in some cases where irrigation is practiced, there is no control on how much water content the soil needs according to the growth stage of the crops. Therefore, there is need to use IoT systems to monitor the water content in the soil so that irrigation is done when needed. The use of IoT would promote proper use of water but also minimizes risks of water lodging due to excessive irrigation.

In this work, a model integrating IoT and ML is proposed. The model uses historical yield data and current weather data to predict crop yield. The IoT system captures real time weather data which is fed into the Machine learning model for prediction purposes. A prototype of the system is developed and deployed on the farm for real-time data capturing and crop yield prediction using a robust prediction model. This system responds directly to the need for data driven approaches to policy making in the agricultural sector to ensure food security, especially in low-to-middle income countries. By capturing real-time weather parameters, the system has potential for use beyond crop prediction as it can also be used to optimize irrigation and water use in Rwanda. Furthermore, by using weather and crop yield data, the model makes it possible to analyze the relationship between crop production and climate change with time. Specifically, the following are the contributions of this work:

- Presents the designing and development of a crop yield prediction system called SMART-CYPS that integrates IoT and machine learning.
- Using collected real-time weather data and historical crop yield data, we analyze Irish-Potato and Maize seasonal yields for Musanze District in Rwanda.

In summary, the contribution this study to the body of knowledge is the integration of IoT and ML to monitor weather and hydrological variability and predict the quantity of the harvest at each growing stage of the crop. Our focus was on those variables that are not controlled by human intervention but can be accurately monitored through IoT technology. We designed and implemented a low-cost IoT device making the system affordable to most small scale farmers' cooperatives in sub-Saharan Africa.

2 Related works

Emerging technologies continue to play a central role in solving different problems, including those related to the basic needs of people, such as food security. Various studies used IoT and artificial intelligence (AI) to provide solutions to the problems related to insufficient crop yield. The optimization of crop production may start with matching of crops to specific weather condition. For instance, the study conducted by Amna et al. [14] used IoT and ML for smart selection of crops to be planted in a specific farm by monitoring different parameters such as rainfall temperature, humidity, nitrogen, potassium, pH, and carbon dioxide (CO_2). Since each crop has different nutrient requirements, the designed system gathers data from the soil, and with ML algorithms, the appropriate type of crop is recommended. Similar studies have been conducted using the same parameters in [15–17].

The quantity and quality of crop production can also be enhanced by predicting crop diseases using IoT and AI. In their study, Ravesa et al. [18] proposed a prediction model for apple orchard diseases using data analytics, and ML in an IoT system. They fed a linear regression model real-time data collected by wireless sensor nodes and IoT nodes. The design consists of several IoT sensor nodes placed across the apple orchard and a nearby gateway for gathering data from the nodes' mesh. A network was created between the nodes, and a nearby gateway. The fog node collects data from a certain number of nodes in the network. For each location, data analysis is performed in real-time to enable prompt responses as necessary. The application might provide farmers with real-time updates on the health of their apple orchards, allowing them to take immediate action against any potential illness like scab if it is accurately and properly forecasted.

On the other hand, optimizing crop yield can be done through monitoring various influencing parameters, such as those related to weather and soil properties. The system designed by [19] monitored daily rainfall and temperature for the prediction of the crop harvest. The architecture of the system consists of a peasant farming layer where the farmer uses a mobile application to input the seasonal crop yield into the system for storage. The same application is used by the farmer to visualize the predicted crop yield as well as weather data that are remotely collected by IoT sensors. The second layer of this system is a wireless sensor network that consists of ultra-low power transceiver modules that are used for communication among the sensor nodes. IoT-based hardware is the third layer of the system. It was made up of a general-purpose microcontroller connected to the transceiver module, which is the global system for mobile (GSM). The last layer was the cloud computing platform, which comprises four cloud computing services: an IoT message hub for enabling two-way communication between system components; data storage service for storing data from sensors; data analytics service for data mining; and a machine learning service for the implementation of predictive analytics for crop production.

An IoT ecosystem for smart agriculture framework proposed by Vu et al. [20] comprises four main components. The first component which consists of IoT devices for monitoring the farming environment, encompasses sensors to gather information such as weather-related data and actuators to enable physical action as a result of decisions based on the information coming from sensors. The second component of this framework is the communication technology, which consists of infrastructure and communication protocols that are responsible for data transfer from the sensor node to the cloud. The third component of this framework is the cloud, which comprises database and server applications responsible for data processing and storage, big data analytics, and decision making. The fourth section is the end user (farmer or system owner) who interacts with the system using an application. Other studies looked at how to anticipate agricultural yields using remote sensing. Those that have been in practice for a while use satellite and drone images for crop harvesting decisions [20–24].

The monitoring of crops on a global and regional scale is currently greatly aided by better new technologies [21]. There are recent studies that focused on the use of ML to predict crop yields. Various ML algorithms (MLA) have been trained using datasets comprising different environmental features and/or soil variables to identify the best performing model. One of the most recent study on the fusion of ML and IoT for precision agriculture was conducted by Arfanul et al. They used a voting regressor (VR) meta-estimator to train a dataset with three boosting regressors as their base models: the gradient boosting regressor, xgboost regressor, and catboost regressor. The experiment showed that VR performed better than other MLAs [25]. Iniyan S et al. also used the strategy of attempting multiple ML models for crop recommendation and yield prediction using environmental and soil parameters. They investigated the performance of different MLAs such as multiple linear regression, decision tree regression, gradient boosting, lasso net, and long short-term memory (LSTM). Their experiment results showed that the most efficient model was LSTM with an accuracy of 86.3%. Additionally, Preeti et al. proposed CNN-Bi-LSTM_CYP, a hybrid approach combining convolutional neural networks and bidirectional LSTM. The proposed hybrid model outperformed traditional stacked-LSTM, ARIMA, and GPR in predicting sugarcane [26].

Similarly, an approach combining deep learning (DL) dimensionality reduction and weight-tuned deep convolutional neural network (WTDCNN) was proposed by Subramaniam et al. [27] achieving maximum accuracy of 98.96% and lower error values of 0.034 MSE and 0.065 FNR. Fatima M. proposed a framework utilizing three ML models (decision tree regressor, random forest regressor, and extra tree regressor) and two statistical models (Pearson's correlation coefficient (PCC) and multiple regression (MR)) for a crop yield prediction algorithm. Another ensemble-based ML technique was proposed by Liyakathunisa [28] who performed predictions in two levels: (i) level-0 used logistic regression (LR), classification and regression trees (CART), support vector machines (SVM), and k-nearest neighbors (KNN), and (ii) level-1 used RF for crop recommendation. An accuracy of 99% was achieved by the proposed ensemble ML technique to predict yield for 22 crop categories. Sundaresan et al. trained a dataset using KNN, DT, RF, GB, and XGB, and based on their performance, KNN was implemented in the crop and fertilizer recommendation system [29].

IoT devices with three ML models were used to predict crop yields in [30]. IoT sensors collected environmental and soil data, and ANN analyzed the sensor data and extracted features, while fuzzy logic and SVM models handled uncertainty in the data and made predictions and classifications, respectively. Pukrongta et al. [31] employed IoT devices and unmanned aerial vehicles (UAVs) to gather environmental data and vegetation imagery respectively. They trained various ML models including catboost, decision trees, elasticnet, gradient boosting, huber, KNN, lasso regression, linear regression, M estimators, passive aggressive, RF, and ridge regression. Performance evaluation using the weighted ensemble prediction (PEnsemble) showed that the PEnsemble 4 methodology provided better accuracy of 91%, improving the prediction of grain yield compared to conventional methods.

As highlighted by the above cited works, IoT and ML are key technologies in the realization of food security. IoT supports real time monitoring of crops and weather parameters, whilst ML algorithms can use both sensed and historical data to predict crop yields. Knowledge of anticipated harvest empowers the farmers to plan ahead, e.g., by searching for markets. Famine early warning is another application of crop production prediction, which allows decision-makers like the government to establish a strategic plan to address the impending food insecurity. However, unlike previous studies, this work focuses on crop yield prediction using a cost effective IoT devices that gather real-time data and ML for predictive analysis. In contrast, recent studies relied on ML algorithms, missing the integration of IoT [17, 18, 29], whereas those using both IoT and ML are expensive [31] or considered IoT data but did not implement it in their system or enter data manually [29]. In the current study, we designed, developed, implemented, and experimented a real-time data driven cro yield prediction system. Considering the impact of climate change on Rwanda's agricultural sector, we believe the developed system will support various initiatives being undertaken by the government to ensure food security in the country.

3 SMART-CYPS—system design and development

3.1 Methodology

Data collection and modelling: This study builds on the findings reported in [32], where we conducted a study that analysed the impact of climate conditions on crop production. As part of the study, historical data about meteorology and crop yields (i.e., for Irish Potato and maize) were collected and using this data modelling was performed to determine the best ML model to be implemented for crop yield prediction. Furthermore, the study identified the optimum values of weather parameters for optimum yields. Therefore, this study uses the values from that study to develop a crop yield prediction system that integrates IoT and ML.

This study was limited to the analysis of weather and hydrological parameters, as these were the data readily available and could also be monitored effectively using IoT sensors. The choice of these parameters was guided by our previous findings in reference [32], which demonstrated the significant potential of weather variables in predicting crop yield. We made the assumption that other human-controllable factors, such as pest infestations, soil nutrients, or crop management practices, were well-managed and thus would not introduce significant variability.

3.2 System architecture

The architecture of the developed crop yield prediction system is shown in Fig. 1. The data used by the system for prediction comprises secondary data and primary data. The secondary data includes, crop yield data and climatic data, i.e., rainfall, temperature and humidity but also crop yield data. The secondary data can be accessed from various databases

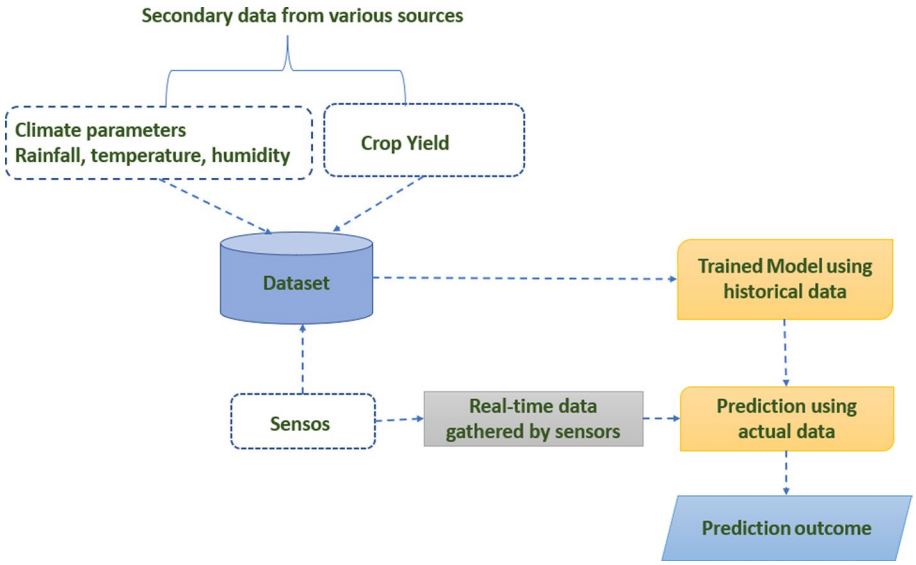


Fig. 1 SMART-CYPS—crop yield prediction system architecture

whilst the primary data is continuously being collected by the IoT sensors and being fed into the system. As can be seen from the Fig. 1, the sensors collect data for two purposes, namely; for prediction and storage. All real-time data is fed into the deployed model for harvest prediction. Informed by the findings in [15], the ML model trained for prediction is the RF regressor. The model is configured in a such way that, it predicts the harvest basing on the crop growing stage or monthly data batches and not basing on single data prediction.

3.3 SMART-CYPS—IoT sub-system

3.3.1 System design and functionality

The IoT sub-system of the SMART-CYPS comprises various sensors that collect data related to climatic conditions including, humidity, temperature, rainfall, and soil moisture. The system further has sensors that monitors system battery voltage which helps in determining the battery status. Other components include, the GSM module used for communication, and GPS module used for locating the node.

Table 1 provides the details of all the electronic components of the IoT sub-system.

All components are mounted on the printed circuit board (PCB) allowing the data collection and processing. Figure 2 shows the circuit design. The powering system is made of three main components: solar panel (20 W), Battery, and charge controller. Figure 3 shows the system prototype including the powering components.

3.3.2 System communication

Figure 4 shows the communication architecture of the SMART-CYPS. As pointed out above, the IoT data are collected using sensors (sensor nodes labelled as Edge Layer in this context) that transmits the data to the cloud through GPRS protocol within the telecommunication cellular network.

The sensors transmit the data to the cloud with associated tokens. The time interval for temperature, humidity, and soil moisture is 30 min, whereas rainfall data is only transmitted when the value is different from zero (i.e., when it is raining); otherwise, the default value for this parameter is 0. To manage the data ingestion from heterogenous devices, the data is sent and received in JavaScript Object Notation (JSON) format. The past production information and weather conditions are also stored in the server where data mining techniques will be used to produce a report showing the future production predicted by the system. To ensure the security of the data, two databases are established to avoid data loss. Prior to model training, data pre-processing is performed by converting the data into comma-separated values (CSV) format and cleaning it by removing anomalies and handling missing data. To ease the computation by avoiding model testing overhead, cloud services are used under the container paradigms as microservices. This scheme proved to be efficient while in Docker environment. For the data that may fail to be transmitted due to some issues such as low battery's power or network desruptions, each sensor node is equipped with an internal data logger that stores the data for later transmission when the network is

Table 1 Description of components used for the development of the CYPS

Component	Description	Function	Operating Voltage
ATMEGA328P-PU	Microcontroller IC	Performs data processing function	1.8–5.5 V
DHT22	Digital Temperature and Humidity Sensor module	Measuring the temperature & humidity of the surrounding air	5V
FST100-2007 RS485 Modbus 24V DC Rain and Snow Sensor	Detects rainwater	Measures whether there is rain or snowfall outdoors	10-30V-DC
Soil moisture sensor	Soil moisture VWC water content test sensor probe with 3 pins	Measuring the soil water content	5-30V-DC
Voltage sensor	Voltage detection module	Monitoring the status voltage saved by the battery powered by the solar panel	5V
GY-NEO6MV2	GPS Module	Detecting the location of the sensor node	3-5V
SIM800L	GSM/GPRS module	Connecting the device to the GSM network	4.2V

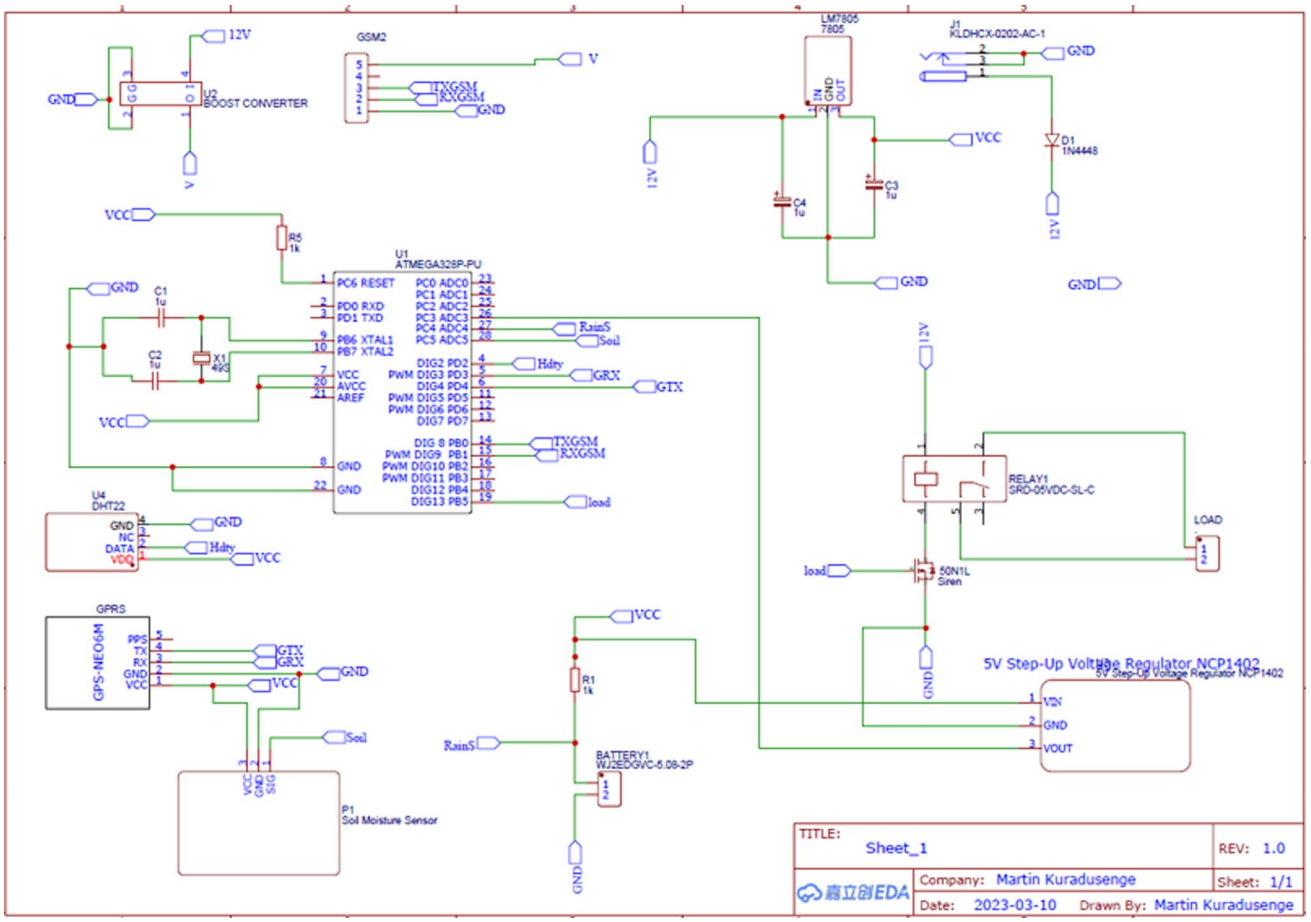


Fig. 2 Circuit diagram

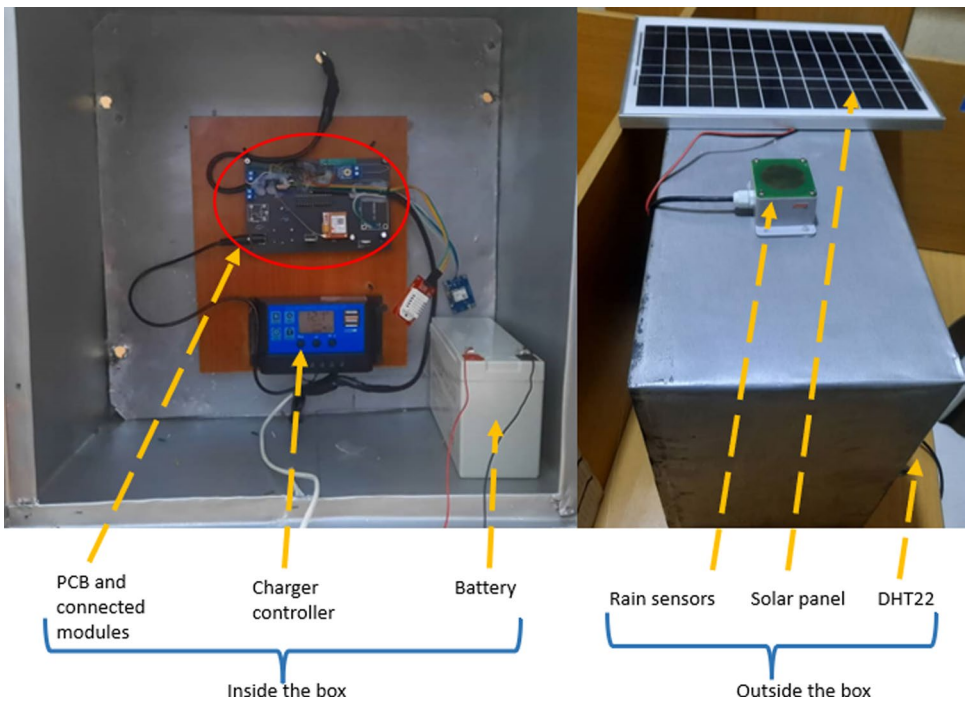


Fig. 3 Prototype system with the powering components

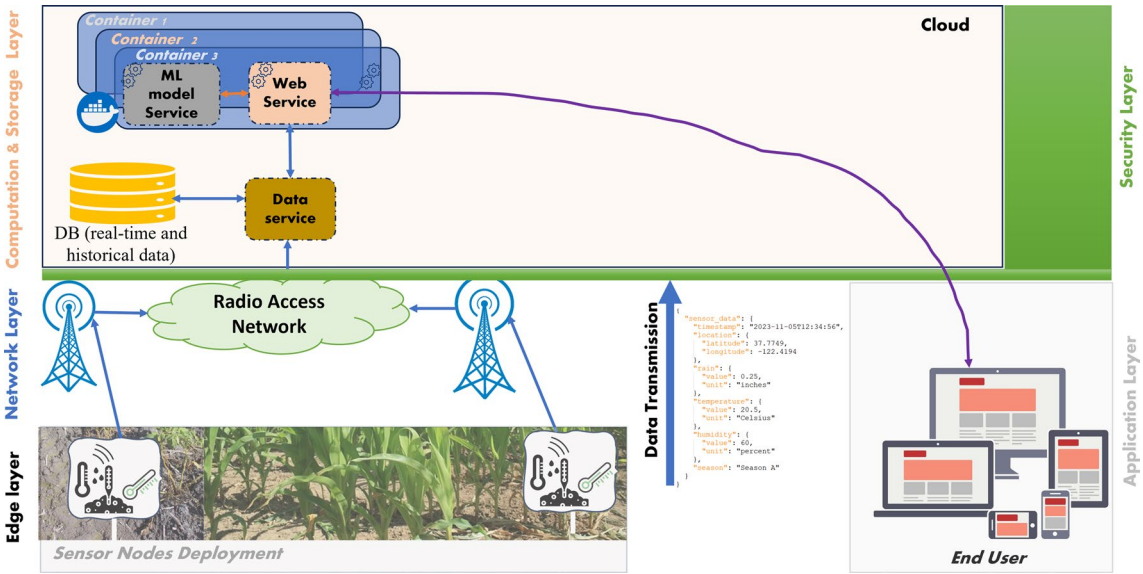


Fig. 4 IoT-based data collection process

recovered. An authenticated farmer/policymaker can be able to interact with the system to see those reports and predictions. The developed web solution can predict the crop yield depending on the current data from the sensor nodes and the historical harvest. This prediction is due to the integrated model in the cloud services.

3.3.3 SMART- CYPS—Configuration, calibration, and validation

Initial calibration was conducted during the testing phase to establish baseline accuracy. To ensure precise and consistent readings, we calibrated all sensors according to the manufacturer's accuracy specifications provided in the data sheet. The calibration accuracies were ± 0.5 °C, ± 2% RH, ± 3%, and ± 10% for air temperature, air humidity, soil moisture, and rainfall sensors respectively. Additionally, we conducted cross-validation by using multiple sensors of each type, comparing their readings to identify any discrepancies. Furthermore, we benchmarked sensor readings against data from nearby weather stations to validate accuracy. This process helped to ensure accuracy and make necessary calibrations. Finally, we planned and performed regular calibration, typically on a quarterly basis, or more frequently if necessary.

In addition to the sensor node calibrations, other settings and configurations were performed to adapt the yield prediction to different crops. Initially, we tested the system on potatoes and maize with their respective datasets, but to further test the generalization of the system, we also tested it on beans. Given that each crop has unique requirements for weather, soil, and management practices at each growing stage, we had to make some adjustments when applying predictions to different crops. These adjustments included using distinct datasets, modifying the growing stages, and setting specific start and end times for predictions (as predictions are made at each growing stage).

3.4 SMART-CYPS—Integrated machine learning model

This work implements the RF regressor based on its best performance in crop yield forecasting as reported in our previous study [15]. RF is the collection of decision tree-based ML algorithm. It is an ensemble of tree predictors where each tree relies on the values of a random vector that is sampled independently and follows the same distribution for all trees in the forest. It is effective for both classification and regression problems [33]. Random forests for regression are constructed by growing trees based on a random vector θ such that the tree predictor h(x,θ) yields numerical values instead of class labels. The output values are numerical, and we assume that the training set is independently drawn from the distribution of the random vector Y,X [34]. The mean-squared generalization error for any numerical predictor h(x) is:

$$h(x) = \boldsymbol{E}_{X,Y}(\boldsymbol{Y} - h(\boldsymbol{X}))^2 \tag{1}$$

The RF predictor is created by calculating the average of k trees {h(x, θk)}.

3.5 SMART-CYPS—machine learning model performance evaluation

We adopt the Mean Absolute Percentage Error (MAPE) as the performance evaluation metric, which is calculated using the following formula:

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{A_i - P_i}{A_i}\right|100 \quad (2)$$

where MAPE is the Mean Absolute Percentage Error, n is the total number of data points (or observations), Ai represents the actual or observed values, and Pi represent the predicted values for the corresponding observations.

3.6 SMART-CYPS—maintenance

To ensure the reliability and longevity of the system, we planned and conducted comprehensive maintenance. Initially, we provided onsite training to system users, including cooperative representatives, agronomists, and local leaders. The training covered how to setup the system, how to access data, record data (such as actual crop yield), enable or disable predictions at the beginning or end of the agricultural season, and perform basic technical support and environmental protection for the IoT devices.

We conduct regular onsite visits to inspect the physical condition and environmental protection of the IoT devices. The databases that store data from sensor nodes are regularly checked to ensure that new data is stored and backed up. Each device is equipped with a voltage sensor to monitor battery life remotely, allowing us to intervene when necessary. To ensure data accuracy and reliability, we regularly review the data generated by sensors in the database and perform calibration as needed. The system administrator should regularly check system logs and take appropriate actions.

4 Results

4.1 Real time data gathering and visualization

The designed and developed system was deployed in the study area after six months of testing. The IoT system collects data on daily basis from the farms and sends it to the cloud for long-term storage and future analysis. Figure 5 is one of an IoT node deployed on the farm.

The system dashboard displays the real time data coming from three of the sensor nodes. In the back end the rest of the data are stored on daily basis. In real-time, the data can be visualized on the system dashboard. The figure also shows the data summary and at the end of the day the minimum, maximum and average data are saved in the database. At



Fig. 5 Deployed sensor node

the end of the season, the rest of the data together with actual crop yield are added to the historical data. As shown in Fig. 6, the system has different modules. Below is a description of each of the module:

Managing users: used to manage users such as adding, removing, and assigning roles.
Cooperative module: used to visualize or add the name of farmers cooperatives that will have access to the information in the system.
Roles: used to add the visualize or add the managing roles.
Devices: module is used to track the device. It provides the GPS coordinates and location on the map. This module also provides the battery's voltage level.
Crop category: provides the option to select or add crop types for yield prediction.
Data visualization: comprises historical data for weather (rainfall, temperature, and humidity), and harvest.

Since the system's deployment, sensors have continuously captured and recorded the data in the database. Figure 7 shows the variation of data over time since the deployment of the system prototype, January 2022 to July 2023.

As illustrated in Fig. 3, the system relies on a solar panel system as the source of energy, a necessity in remote areas without access to the main power grid. The web application is used to keep tabs on the status of the battery. Figure 8 portrays the voltage variations for the period of 1 year and half. The figure illustrates a decrease in voltage during the rainy seasons in Rwanda, spanning from March to May and September to December. It was common to observe interruptions in data transmission during these periods.

4.2 Correlation of weather parameters

To identify the correlation among the weather variables, we utilized the data collected by the system throughout the study's duration. The correlation matrix depicted in Fig. 9, reveals some significant correlations. Specifically, there is a moderate positive correlation of 0.51 between rainfall and soil moisture, a moderate positive correlation of 0.41 between rainfall and humidity, and a small negative correlation of—0.25 between rainfall and maximum temperature. Furthermore, maximum temperature exhibits a moderate negative correlation of—0.36 with soil moisture, and a moderate negative correlation of—0.58 between humidity and maximum temperature. Additionally, a positive correlation of 0.48 was observed between soil moisture and humidity.

In addition, the sensitivity analysis indicated the impact of weather variables used in this study on crop yield. As shown in Table 2, the most significant variable, which has a direct impact on crop yield, is rainfall with a sensitivity index of 0.344, followed by soil moisture, which has a moderate impact on crop yield with an index of 0.097. Minimum and maximum temperatures have less contribution to crop yield, with indices of 0.017 and 0.022, respectively.

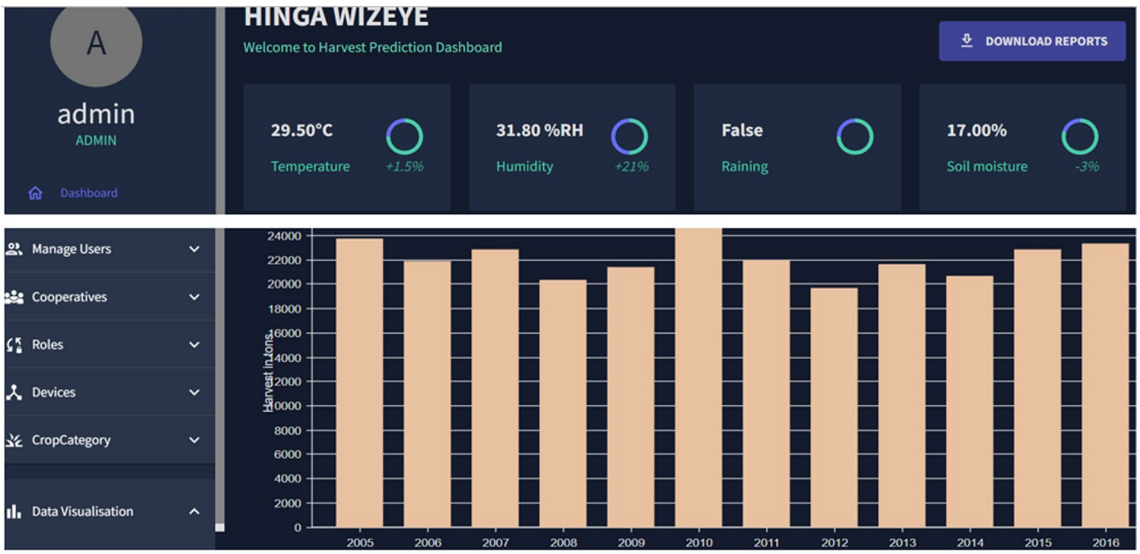


Fig. 6 Real time data visualization. The left frame comprises different modules of the web portal, the top frame shows real-time data, and the middle frame visualizes the data in graph form

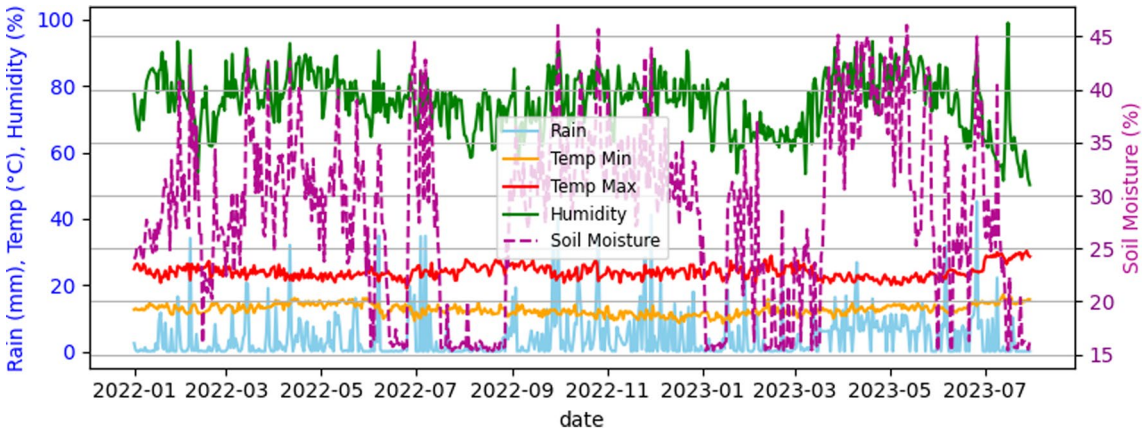


Fig. 7 Weather data captured by one of the deployed sensor nodes. The left Y-Axis represents rainfall (in mm), temperature (min and max in °C), and relative humidity (in %). The right Y-Axis shows soil moisture (in %). The minimum recorded values are as follows: 0 mm for rainfall, 8.4 °C as the lowest temperature, 18.9 °C as the highest temperature, 50.2% as the relative humidity, and 15.3% as the soil moisture. The maximum recorded values are 45.15 mm for rainfall, 17.1 °C as the lowest temperature, 30.3 °C as the highest temperature, 99% as the relative humidity, and 46.27% for soil moisture

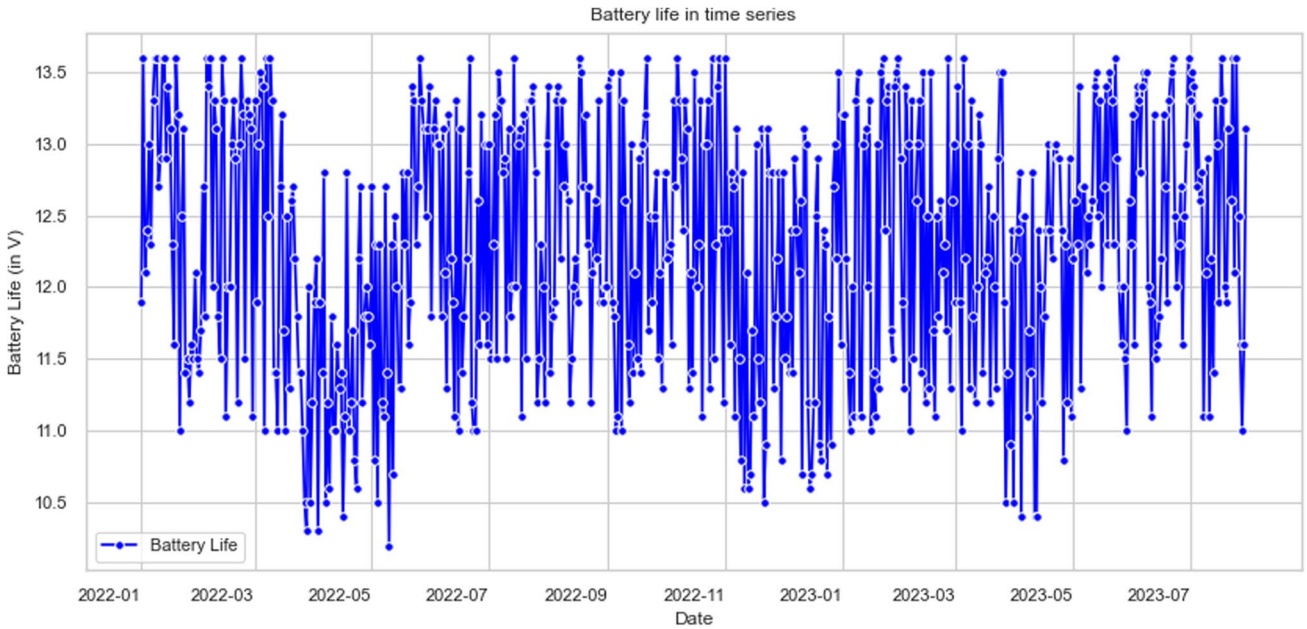


Fig. 8 Battery life data from the voltage sensor

Fig. 9 Correlation among weather parameters

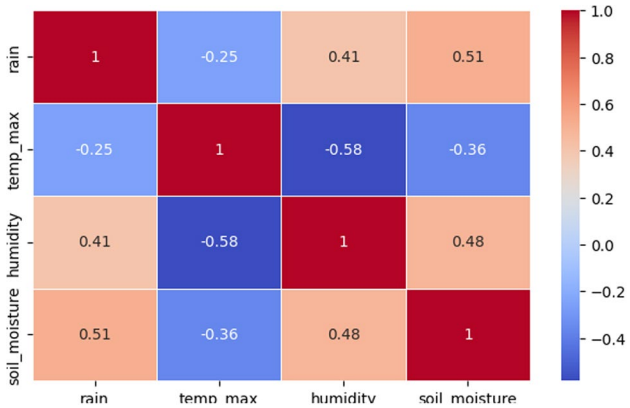


Table 2 Sensitivy first and total order

Sensitivity type	Variables				
	Rainfall	Max temp	Min temp	Humidity	Soil moisture
First order	0.344	0.022	0.017	0.066	0.097
Total order	0.378	0.018	0.032	0.496	0.530

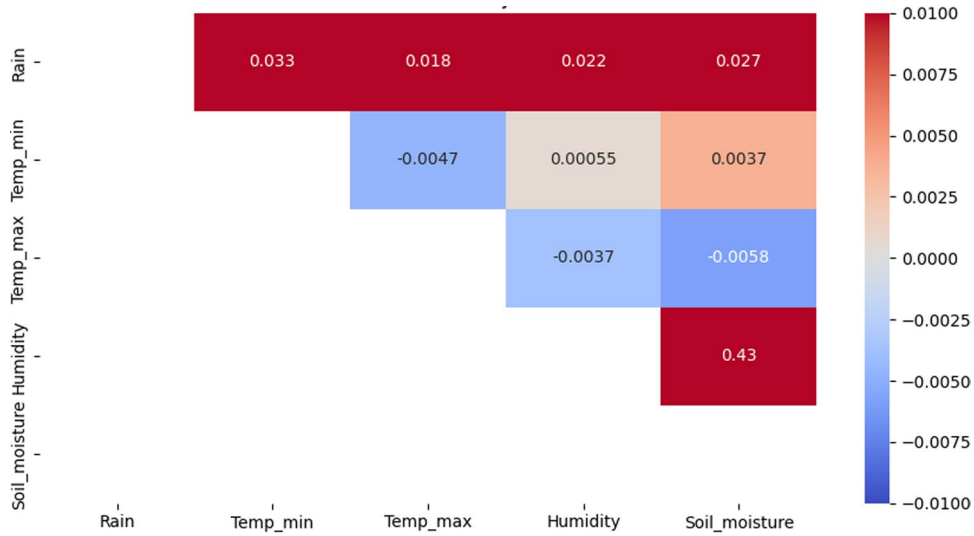


Fig. 10 Sensitivity second order indices

Considering the total order sensitivity, soil moisture, along with its interaction with other variables, is the most contributing factor to crop yield with a sensitivity index of 0.53. Humidity and rainfall, in interaction with other weather variables, play crucial roles in crop yield with sensitivity indices of 0.496 and 0.378, respectively.

Second order sensitivity is shown in Fig. 10. The value of 0.0329 indicates a positive interaction effect, suggesting that the combination of rain and minimum temperature significantly influences the yield. On the other hand, the sensitivity index of 0.43 for the interaction involving humidity and soil moisture is substantial, showing a strong interaction effect, meaning these two variables together have a significant impact on the predicted yield. Conversely, the negative values for the interactions involving maximum temperature and soil moisture might suggest that their combined effect is less straightforward or possibly reduces the effect of one another on the yield.

In brief, for the main effects, rain and soil moisture have the most significant direct effects on the predicted yield. In contrast, soil moisture and humidity have the highest overall impact when considering interactions with other variables for the total effects, playing a critical role in the predicted yield. Since the soil moisture highly depends on the rainfall, the latter has an important impact on the crop yield as well.

4.3 Prediction results and actual yield for irish potatoes and maize

Currently, SMART-CYPS monitors the cultivation of two crops: Irish potatoes and maize, which are rotated between seasons A and B in the study area. Typically, maize is planted during season A, and Irish potatoes are cultivated in season B. Since the system's deployment, we have monitored one season of maize and two seasons of Irish potatoes. In this area, maize has a growth period of six months, while Irish potatoes are harvested after four months. The subsequent tables Tables 3, 4, and 5) present the predicted and actual production for each crop during three agricultural seasons. The system forecasts crop yields at various growth stages.

The growing stages considered for Irish potatoes are sprout development, vegetative, tuber initiation and bulking, and maturity [35]. The initial prediction was conducted at the end of the sprout development phase of Irish potatoes, while the last occurred during maturity. As shown in the Table 3, the actual yield for Irish potatoes in the 2022 season B was 8717 kg ha-1, whereas the predicted yield ranged from 11,373 kg ha-1 to 12,109 kg ha-1. The prediction error (MAPE) for this season was 0.339. In the agricultural season B of 2023, the crop yield was 9837 kg ha-1, and the predicted yield varied from 11,985 kg ha-1 to 13,529 kg ha-1, with a prediction error of 0. 309.

Table 3 Comparison of the yield predicted by CYPS and the actual yield for Irish potatoes (Season B 2022)

Prediction time (growing stage)	Predicted yield (kg ha-1)	Actual yield (kg ha-1)	Accuracy (MAPE)
21 days (sprout development)	12,109	8717	0.339
2nd month (vegetative & tuber initiation)	11,373		
3rd month (tuber bulking/filling)	11,639		
4th month (maturity)	11,594		

Table 4 Comparison of the yield predicted by CYPS and the actual yield for Irish potatoes (Season B 2023)

Prediction time (growing stage)	Predicted yield (kg ha-1)	Actual yield (kg ha-1)	Accuracy (MAPE)
21 days ADP (Sprout development)	13,529	9837	0.309
2nd month (vegetative & tuber initiation)	12,898		
3rd month (tuber bulking/filling)	11,985		
4th month (maturity)	13,130		

Table 5 Comparison of the yield predicted by CYPS and the actual yield for maize (Season A 2023)

Prediction time (growing stage)	Predicted yield (kg ha-1)	Actual yield (kg ha-1)	Accuracy (MAPE)
1st month (establishment & vegetative)	1758	1837	0.177
2nd month (vegetative)	1222		
3rd month (tasseling)	1580		
4th month (cob setting)	1345		
5–6th month (grain-filling & maturity)	1651		

Six growing stages were considered to predict maize production. Those are establishment, vegetative, tasseling, cob setting, grain-filling, and maturity [36].

The results in Table 5 are for agriculture season A of 2023 that started in September 2022. The predicted maize yield varied between 1222 kg ha-1 and 1758 kg ha-1 whereas the actual maize yield was 1837 kg ha-1. The performance of the prediction was 0.177 MAPE. In addition to maize and potatoes, the system was also tested on beans to evaluate its scalability, as discussed in the next section.

5 Discussion

Crop yield is a consequence of a range of parameters, which can be broadly classified into two categories: factors within human control and those influenced by natural forces. Human-controllable factors encompass various elements, including the choice of crops or seed variety tailored to the specific region, planting schedules, soil preparation, the application of fertilizers, irrigation methods, weed management, and the use of pesticides, among others. In the study area, these variables fall under the supervision of local agronomists who collaborate closely with farmers, offering guidance and support as needed. Therefore, the natural factors affecting crop yields are the subject of monitoring in this study. Weather data is collected for two primary purposes: predicting crop yields and creating a historical record for future reference. Predictions are conducted monthly, aligning with the various growth stages of the crops, as indicated in Table 3 through Table 5. The prediction process commences following planting and is initiated by the responsible individual, such as the agronomist or the leader of the farmers' cooperative.

5.1 Rainfall and soil moisture

The SMART-CYPS incorporates a rain sensor designed for rain detection. This sensor furnishes binary values (0 or 1) to signify the presence or absence of rain. The preference for a rain sensor over a rain gauge was driven by the unavailability

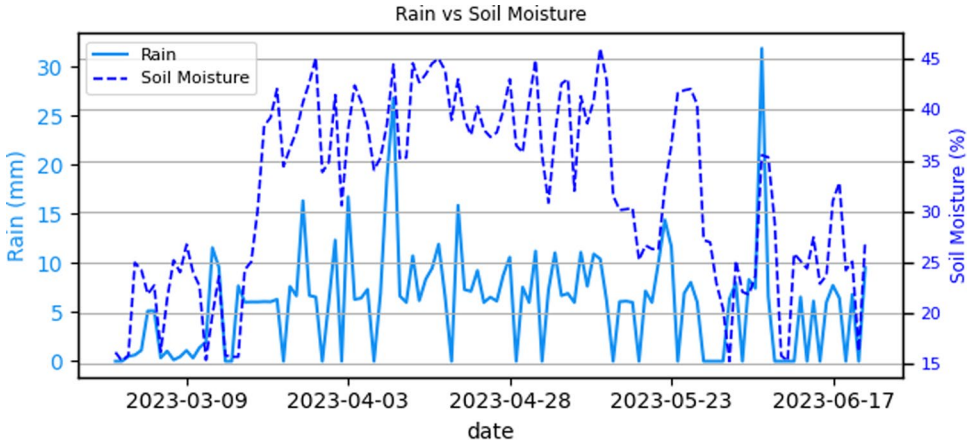


Fig. 11 Correlation among weather parameters

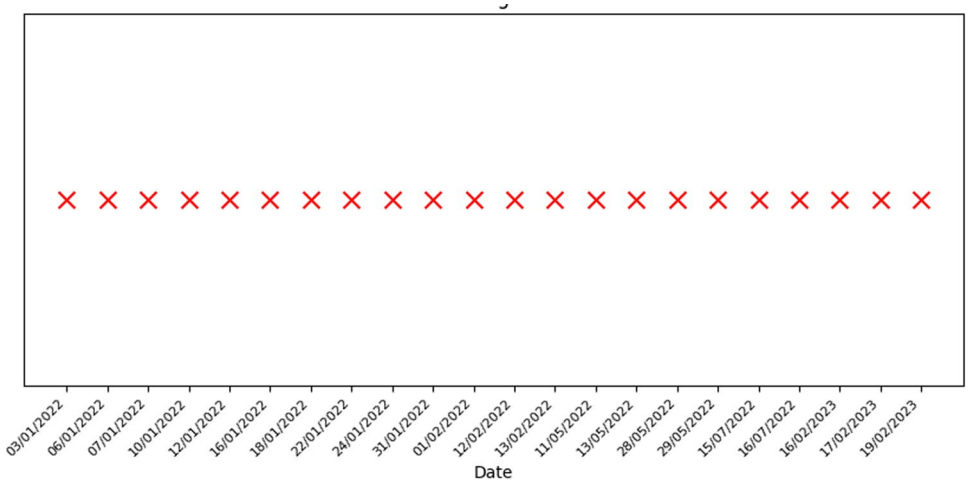


Fig. 12 Inconsistences between rainfall data from rain gauge and rain sensor

of a rain gauge equipped with network capabilities for real-time rainfall data acquisition. To overcome this limitation, we adopted a strategy that involved utilizing rain and soil moisture sensors. This approach enables us to observe shifts in soil moisture levels during rainfall events and supplement this data with secondary rainfall information obtained from the meteorological office. During consecutive rainy days, the soil moisture increases even when the amount of rainfall is not substantial. This indicates that soil moisture depends not only on the daily rainfall but also on antecedent rainfall. The correlation between rainfall and soil moisture is shown in Fig. 11

5.2 Rainfall data from rain gauge vs rain detected by sensor node

Upon comparing the rainfall data obtained from the rain gauge with that from the rain sensor, we discovered discrepancies between the two sources. For example, there were some instances where the system signaled rain, but the recorded rainfall data was zero (0). Contrariwise, there were dates when the rainfall data exceeded zero, yet the system did not detect any rain. These inconsistencies are depicted in Fig. 12.

The inconsistency is due to the locations of the device versus the location of the rain gauge. That means there are some cases where rain does not reach the entire area and this shows the need of increasing the distribution of rain gauges for more accurate information about rainfall in the study region.

5.3 Prediction performance

During every growth stage, the SMART-CYPS computes yield forecasts using cumulative rainfall, average minimum temperature, and average maximum temperature from the start of that stage. It is important to note that humidity and soil moisture were excluded due to insufficient historical data. These parameters will be incorporated into the system once

enough data is recorded. Tables 3, 4 and 5 demonstrates that predicted data varied across growth stages in response to weather conditions, while actual data remained consistent since it is recorded at the conclusion of the agricultural season. The results in Table 3 indicate an accurate prediction of Irish potato yields with a precision of 0.339, which is a commendable outcome. A similar level of model performance was maintained in the subsequent seasons, with a MAPE of 0.309 for predicting the yield of the same crop.

The maize yield predictions for the 2023 agricultural season A demonstrated a significantly improved performance, as indicated by a MAPE of 0.177. Predictions were conducted monthly, a total of five times, utilizing the same parameters as those applied to Irish potatoes. In our study, maize is typically harvested 6 months after planting. During the initial five months, we conducted predictions because during the sixth month, the maize crop is no longer developing, as it is intended for grain drying purposes. The higher frequency of maize yield predictions might be attributed to the enhanced performance compared to Irish potatoes, which could also be attributed to the data quality used in the prediction process.

5.4 SMART-CYPS—scalability

The disparities between the forecasted and real crop yields may stem from additional factors, such as fertilizer usage, planting timing, pest infestations, soil nutrients, crop management practices and more. Consequently, enhancing the prediction capability may be achieved by incorporating these variables in future research. The study did not include all possible weather parameters and other inputs due to the limited data availability. However, the system's performance remains accurate and robust even if the databases for these parameters and past crop yields are replaced with those used in the current study. This was verified by deploying one of the IoT devices in a different location to predict bean yields. For this test, we utilized a separate dataset for weather, soil, and harvest data. The experiment was conducted during the agricultural season A, 2024, which spans from September to January. The MAPE of 0.142 presented in Table 6 indicates that the system still performs well when changing the environment and datasets.

Currently, the system underwent testing with Irish potatoes, maize, and beans demonstrating its adaptability to different crops. To expand its applicability, customization for various other crops is necessary. Scaling up the project involves monitoring additional parameters including soil nutrients and weather conditions. It is also important to note that while the model is adaptable, some calibration or adjustment may be necessary to account for the specific weather and soil requirements of different crops. Incorporating industrial IoT devices will further enhance data accuracy and prediction capabilities.

5.5 SMART-CYPS Practical implementation

The practical implementation of SMART-CYPS involves the use of IoT sensors for real-time data collection, which integrates seamlessly with existing agricultural practices. The cost of IoT sensors has become increasingly affordable due to advances in technology and mass production. This makes the system more accessible to smallholder farmers, especially when organized in cooperatives (for example the case of Rwanda), which can share the costs and benefits.

Integration with current agricultural practices is also facilitated by the modular design of the system, allowing farmers to start with basic weather and soil monitoring and gradually add more sensors and data types as needed. Additionally, industrial IoT devices can further enhance data accuracy and prediction capabilities, which can be crucial for large-scale commercial farming operations. The system is designed to be scalable and adaptable, ensuring that it can be customized to meet the specific needs of different crops and farming practices, thus making it a viable solution for a wide range of agricultural applications.

Table 6 Comparison of the yield predicted by CYPS and the actual yield for beans (Season A 2024)

Prediction time (growing stage)	Predicted yield (kg ha-1)	Actual yield (kg ha-1)	Accuracy (MAPE)
1st month	644	977	0.142
2nd month	991		
3rd month	798		
4th month	1155		
5th month	1034		

Table 7 Comparison of SMART-CYPS and other studies

Study reference	Type of crop yield is predicted	ML model (s)	Parameters	Data collection method (manual, IoT device, real-time, historic, etc.)	Evaluation technique	Performance	Implemented and tested on-site? Y/N
[31]	Maize	PEnsemble 4 model	NPK, pH, soil temp, soil humidity, EC, water feed, Uv radiation, Evapotranspiration, air humidity, air temp, wind speed	Historic and UAV	Accuracy	0.91	Y
[37]	Cassava, maize, potatoes, sorghum, rice, soybeans, sweet potatoes, wheat, yams	DT Regressor, RF Regressor, ExtraTree Regressor	Temp, precipitation, pesticide,	Historic	Accuracy	0.9814, 0.9903, 0.9933	N
[27]	–	Deep convolutional neural network	Temp, precipitation, pesticide,	Historic	Accuracy, precision	Accuracy:98.96%, precision: 98.67%	N
[38]	–	Weight-based long-term memory	Temp, precipitation, pesticide,	Historic	Accuracy	95%	N
[28]	22 crops categories	SVM, LR, KNN, CART, RF	Temp, precipitation, pesticide,	Historic	Accuracy, precision, recall, f-measure	Accuracy: 99%	N
The current study	Maize, Potatoes, Beans	RF Regressor	Temp, humidity, rainfall, soil moisture	Historic and real time	MAPE	Potato: 0.339 Maize: 0.177	N

Table 8 Threats to validity

Category	Threat	Mitigation technique
Sensor data quality	Inaccurate data captured by sensors may affect the predictions	Sensor nodes are regularly calibrated, and we plan to use industrial sensors during the project scale up
Data availability and completeness	Missing and incomplete data may lead to inaccurate prediction	Provision of redundant devices for robust data collection and implement data pre-processing technique before prediction
ML Model generalizability	ML models might be overfitted to some conditions or sites and may not generalise well to other conditions or locations	We trained different models with different datasets and selected the best for integration
Environmental variability	Environmental factors, such as extreme weather events, might not be captured adequately in the model	In the course of training models, we used different datasets from various locations. Some of them might have data for extreme weather events
Human error data handling	Manual errors in data entry or handling can lead to biases or inaccuracies	We minimized manual data entry. Only the actual crop yield should be recorded manually. Out of range entry will not be accepted
System scalability	The system may face challenges when scaled to larger areas, different crops or agricultural environments without significant customization	We tested the system in different locations, using different datasets with different agricultural environments. In our future work, we plan to develop modular and customizable components
External data dependency	The system relies on external data sources, such as historical weather and yield data, which may not always be reliable or timely	The system captures real time data to be added to the existing. Stakeholders are trained to enter actual harvest data in their area. All data are backed up
Technology limitations	Limitations in IoT technology, battery and connectivity issues may affect data collection	Devices are powered by solar panel, and in one site we duplicated devices. To scale up, we plan to use industrial IoT devices which are more stable
Economic constraints	For some smallholder farmers, budget limitations and operational Constraints might restrict the full implementation and maintenance of the system	We worked with the farmers in cooperatives. The responsibility and the budget for devices, operation and maintenance will be shared. Cooperatives can afford the cost of the device
User adoption and training	Limitations in technological skills for stakeholders (farmers, agronomists, others) who may face challenges in adopting and effectively using the system	Before the lunch of the system, we conducted a training session to the cooperative representatives so that they can also train their respective farmers. Basic technical aspects were provided for some technical and maintenance support

5.6 SMART-CYPS in the context of other studies

In this section, we benchmarked the performance of SMART-CYPS against other studies employing various techniques and parameters. Table 7 provides a comparison of our study with recent ones that conducted crop yield prediction, highlighting their technical aspects and performances.

Most of the studies in Table 7 were limited to modeling or simulation, rather than application in real-world scenarios or direct experimentation. These studies face challenges such as limited realism, validation difficulties, and unaccounted for environmental factors, making it hard to fully capture real-world conditions and accurately predict outcomes without field testing. Only one study implemented data collection using UAVs [31], which can be expensive and challenging for farmers to manage, maintain, and operate. In contrast, the SMART-CYPS proposed in this study is cost-effective for farmers organized in cooperatives and can be easily managed by them, even with basic technical knowledge.

5.7 SMART-CYPS—threats to validity

In this section, we present potential sources of errors or bias that might affect the validity of our research findings. Table 8 summarizes potential threats and their respective mitigation techniques.

6 Conclusions

Predicting crop yields can significantly contribute to reducing food insecurity since such information empowers farmers and other agricultural stakeholders to take appropriate steps well before harvest. In this work, two sources of data have been used to predict crop yields, i.e., real time information provided by the IoT sensor nodes and historical meteorological and crop yields data. The developed SMART-CYPS system has been deployed in Maize and Irish potato farms, however, the system can be adapted for other crop types subject to availability of historical data on climate and crop yields, along with comprehensive information on the crop's weather requirements. This study contributed to precision agriculture by utilizing affordable IoT devices for farmers in small cooperatives, such as those in sub-Saharan Africa. In future endeavors, the SMART-CYPS will be further improved by incorporating real-time rain gauge data and expanding its parameters to include soil properties, fertilizer usage, and more.

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Data availability All data used in this study are accessible online via the portal of the African Centre of Excellence in IoT (ACEIoT) at the University of Rwanda. The data link can be found in the reference section [39]: https://aceiot.ur.ac.rw/SMART-CYPS_data.zip.

Code availability Not applicable.

Declarations

Competing interests The authors declare no competing interests.

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